

Landmark Symmetry and Cognitive Map Formation in Predictive Recurrent Neural Networks

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Abstract

Levenstein et al. (2024) grounded spatial map formation in predictive recurrent networks on the assumption of globally unique observations. We systematically relax this assumption by training identical networks in arenas with graded rotational symmetry (C4, C2, asymmetric) matched for observational information content ($ODI = 0.336$), characterising the representational consequences of observation aliasing across symmetry conditions. We introduced three metrics beyond sRSA: PAA detects global orientational degeneracy, RA detects unit-level aliasing, and C2 Contrast detects group structure encoding. Head-direction input anchors global orientation through transition statistics, preventing degeneracy, but local precision degrades with symmetry order. RA scales as $S4 = 0.223 > S2 = 0.116 > S1 = 0.098$; decoding error worsens from $S1 = 0.365$ to $S4 = 0.524$; distance geometry shifts toward path-dependence ($DTG_{S4} = -0.052$, $DTG_{S1} = -0.020$). These metrics explain multi-field place cell emergence in symmetric arenas, predict head-direction lesion deficits, and reveal why sRSA misses map distortion under partial observation symmetry.

Introduction

Levenstein et al. (2024) showed that a recurrent network trained on sequential predictive loss spontaneously develops spatially structured representations—but validated this exclusively in an L-shaped arena chosen to ensure every position has a unique visual fingerprint. Most rodent electrophysiology experiments use square arenas. Square arenas are rotationally symmetric. The condition Levenstein’s validation depends on—global observational discriminability—does not hold there. We ask what the Levenstein framework actually does under these conditions. Not what the hippocampus does. What the model does. The answer has direct consequences for how the framework should be used, what its representations are grounded in, and what any extension to more naturalistic environments should expect to find.

The head-direction (HD) system tracks global orientation through path integration while landmark anchoring corrects drift via visual cues [5, 1], but symmetric landmarks force these two mechanisms into direct conflict: the same landmark view appears at multiple rotations, so the HD state and the landmark signal become discordant. The double-rotation literature documents that place cells increase their number of firing fields in symmetric arenas [3, 1], with multi-field fraction scaling with symmetry order, and these multi-field patterns emerge spontaneously in the hippocampus without explicit instruction [2].

This paper asks whether landmark symmetry causes global map degeneracy (where different seeds learn incompatible global orientations), local representational folding (where symmetric positions collapse within a single run), or both. A standard metric (seed-to-seed RSA) cannot answer this question because row permutations preserve pairwise distances, making it blind to both global rotations and local folding. We introduce three metrics that separate identifiability from entanglement: PAA and RA for global vs local, C2 Contrast for group structure, and use them to test predictive networks trained on arenas with graded rotational symmetry. We show that head-direction input prevents global degeneracy while local precision degrades monotonically with symmetry order, and that this partial-folding pattern explains multi-field place cell emergence, predicts specific lesion deficits, and uncovers why sRSA fails under observation aliasing.

The Successor Representation framework [4] predicts that spatial representations should be anchored by state transitions. Under observation aliasing, visual inputs are non-injective but transitions between states remain globally distinguishable if movement is controlled: from any state, different actions lead to different future states with different visual trajectories. Critically, head direction provides an unambiguous anchor: high-dimensional proprioceptive information about self-motion accumulates to resolve which alias of the current visual state the agent occupies. We hypothesise that HD-tagged transitions furnish the representational information needed to decode global orientation despite symmetric observations.

This work uses the Levenstein et al. pRNN architecture without modification. The model, training objective, and hyperparameters are identical to their published implementation. The only thing we changed is the environment. Any finding is therefore attributable to the

symmetry manipulation, not to architectural differences. The experiments were designed as a direct stress-test of the framework under conditions it was never validated in—not as a new model of hippocampal function.

Methods

The arena is an 18×18 MiniGrid with 324 positions. All conditions are matched on observational discriminability:

$$\text{ODI} = 1 - \frac{H(X | O)}{H(X)} = 0.336.$$

S4 has C4-equivalent observations under 90° rotation ($H_2 = 4.0$), S2 has C2-equivalent observations under 180° rotation ($H_2 = 2.0$), and S1 has no designed equivalence ($H_2 = 1.0$). The number of seeds is S4 $n = 5$, S2 $n = 3$, S1 $n = 3$.

The model is a NormReLU pRNN with $N = 500$ units. At time t , the hidden state updates as:

$$h_t = \text{NormReLU}(W_h h_{t-1} + W_o o_t + W_a a_t + W_d d_t),$$

where o_t is the egocentric observation, a_t is the action, and d_t is a 12-bin one-hot head-direction input. Training minimises multi-step prediction loss for $k = 5$ steps:

$$\mathcal{L} = \sum_{j=1}^k \|\hat{o}_{t+j} - o_{t+j}\|_2^2,$$

over 80,000 training steps. The three arena layouts that enforce different symmetry constraints are shown in Figure 1.

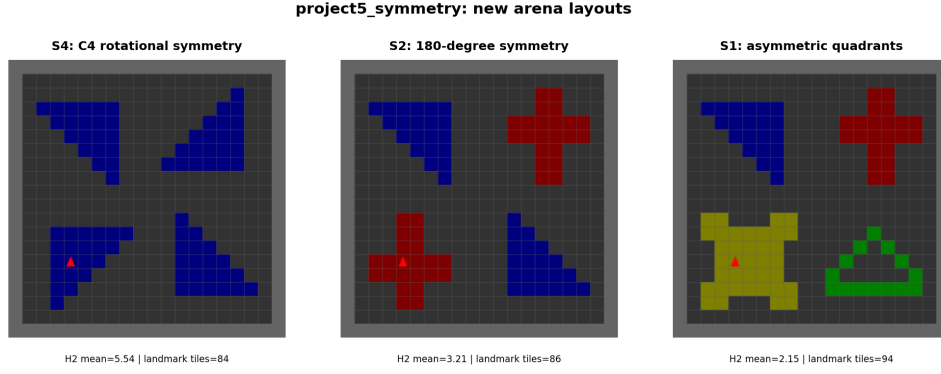


Figure 1: Arena layouts by symmetry order. S4 has C_4 rotational symmetry; S2 has C_2 symmetry; S1 is asymmetric.

The conditions maintain equal observational information content ($ODI = 0.336$) while varying only symmetry. Four metrics quantify the effects. RA (Rotational Autocorrelation) is $RA_u = r(T_u, \text{rot}_{90}(T_u))$ averaged over units with $EVS > 0.05$. PAA (Permutation-Aware Alignment) gain is the best-fit rotated cross-seed RSA minus identity-alignment RSA; values below 0.05 indicate no global orientational degeneracy. C2 Contrast is:

$$\text{C2 Contrast} = \frac{1}{2} \sum_{\text{C2 pairs}} d(Q_i, Q_j) - \frac{1}{2} \sum_{\text{C4 pairs}} d(Q_k, Q_\ell),$$

where C2 pairs are 180° -related positions and C4 pairs are 90° -related. SCI (Symmetry Collapse Index) is $SCI = \bar{d}_{\text{sym}} / \bar{d}_{\text{rand}}$, the ratio of mean distance among symmetric position pairs to mean distance among random pairs in ℓ_2 -normalised hidden space.

The Levenstein codebase was inherited and extended with focus on training efficiency. To accommodate available hardware (RTX 3060, 6GB VRAM), gradient accumulation with `ACCUM_STEPS=8` served as a mathematically equivalent substitute for batch parallelism. The `pRNN_th` rollout architecture was used with $k = 5$ and sequence lengths $T = 200$ or $T = 300$. A critical bug fix involved restoring `squeeze(0)` in the `_forward_theta` implementation, which had been inadvertently removed; this `squeeze` operation had been collapsing the batch dimension into the theta dimension, producing flat loss and collapsed sRSA. Restoring it with gradient accumulation resolved this issue. Training used `torch.compile` and `F.layer_norm` for throughput improvement. The optimizer was RMSProp with $\alpha = 0.95$ and $\epsilon = 1 \times 10^{-9}$, consistent with Levenstein et al., chosen because per-parameter learning rate scaling in this recurrent architecture already performs much of Adam’s adaptive work, making RMSProp the more stable choice for recurrent training on temporally correlated sequences. A total of 80,000 training steps were run per seed across 11 seeds (S4 $n = 5$, S2 $n = 3$, S1 $n = 3$).

sRSA cannot detect the failure modes this paper measures

Seed-to-seed RSA correlates pairwise neural distances with pairwise spatial distances. However, this metric has a critical vulnerability: row permutations of the neural population vector preserve all pairwise distances, so rotated maps yield identical sRSA values even though they represent opposite global orientations.

Figure 2 demonstrates this blindness: two fundamentally different failure modes—global rotation versus local folding—yield identical sRSA despite producing qualitatively different neural codes.

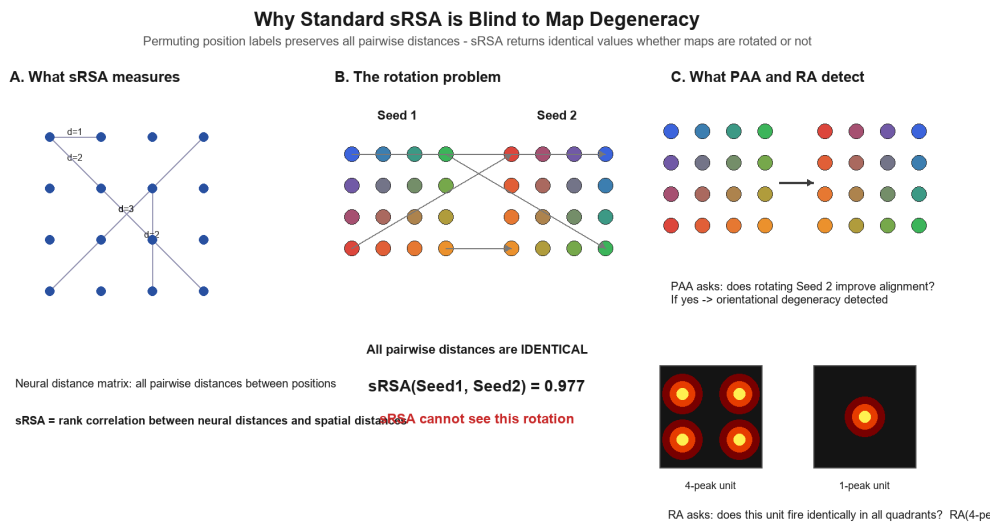


Figure 2: Why sRSA is invariant to permutations. Left: rotated map preserves distances. Right: locally folded map also preserves rank order. Both yield identical pairwise distances.

Three purpose-built metrics address this blindness. PAA detects when rotated alignment between seeds beats identity alignment, revealing orientational divergence. RA measures whether units respond identically at rotationally equivalent positions, capturing unit-level aliasing. C2 Contrast tests whether the population code encodes group structure. PAA, RA, and C2 Contrast separate identifiability, entanglement, and population structure—what sRSA cannot do.

Three possible outcomes frame every result that follows

The space of possible outcomes is structured by two independent questions: Does global orientation remain stable across seeds? Does local precision stay high? This produces three mechanistically distinct failure modes: (1) global degeneracy and local collapse, (2) global stability but local folding (the case we observe), or (3) full robustness.

Figure 3 shows all three outcomes schematised by arena quadrant.

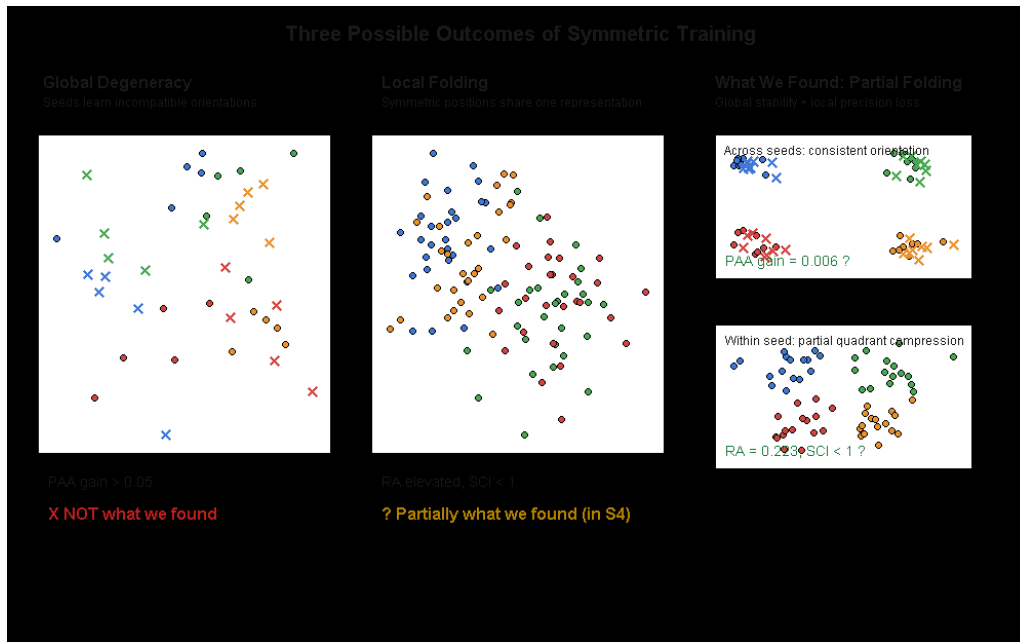


Figure 3: Three mechanistically distinct outcomes. Left: global degeneracy with full collapse. Center: partial folding (observed here). Right: null case (no cost). Quadrant colors show local spatial code.

All three arenas learn spatial representations. S1 (asymmetric) achieves the highest final sRSA despite having the most structure—a pattern sRSA cannot explain. Training loss trajectories confirm convergence across conditions, while the metric dashboard (Figure 4) reveals the consistent pattern across conditions.

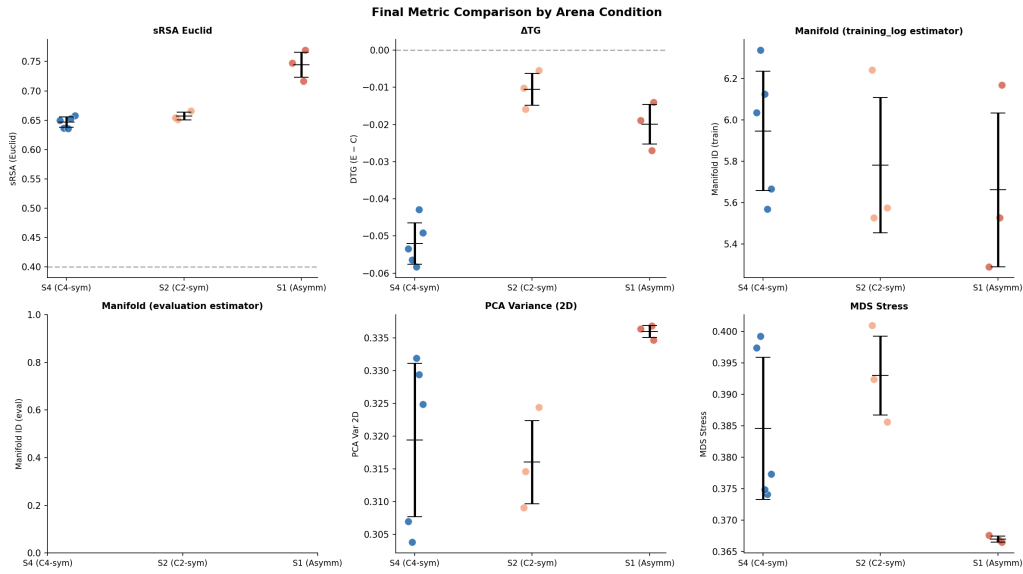


Figure 4: Final metric comparison by condition. PAA (global alignment), RA (unit aliasing), SCI (population compression), and C2 Contrast (subgroup encoding). Points show individual seeds; bars show mean $\pm$ spread.

These four metrics reveal the pattern: global structure is stable (PAA < 0.05), but local precision degrades with symmetry (RA increases with $|C_n|$), and the population code encodes the observation symmetry group itself (C2 Contrast positive in S2, negative in S4). The next page quantifies global orientational stability in detail.

HD input prevents global orientational degeneracy

Global map degeneracy did not occur. Cross-seed RSA alignment is S4 0.977 ± 0.002 , S2 0.975 ± 0.003 , S1 0.974 ± 0.001 . PAA gain is S4 0.010 ± 0.002 , S2 0.008 ± 0.002 , S1 0.008 ± 0.004 with $p = 0.786$ and $r = -0.200$ —indistinguishable across conditions and all below 0.05.

Figure 5 confirms that the primary subspace aligns consistently across all symmetry conditions, suggesting that head-direction input provides sufficient orientational constraint despite landmark symmetry.

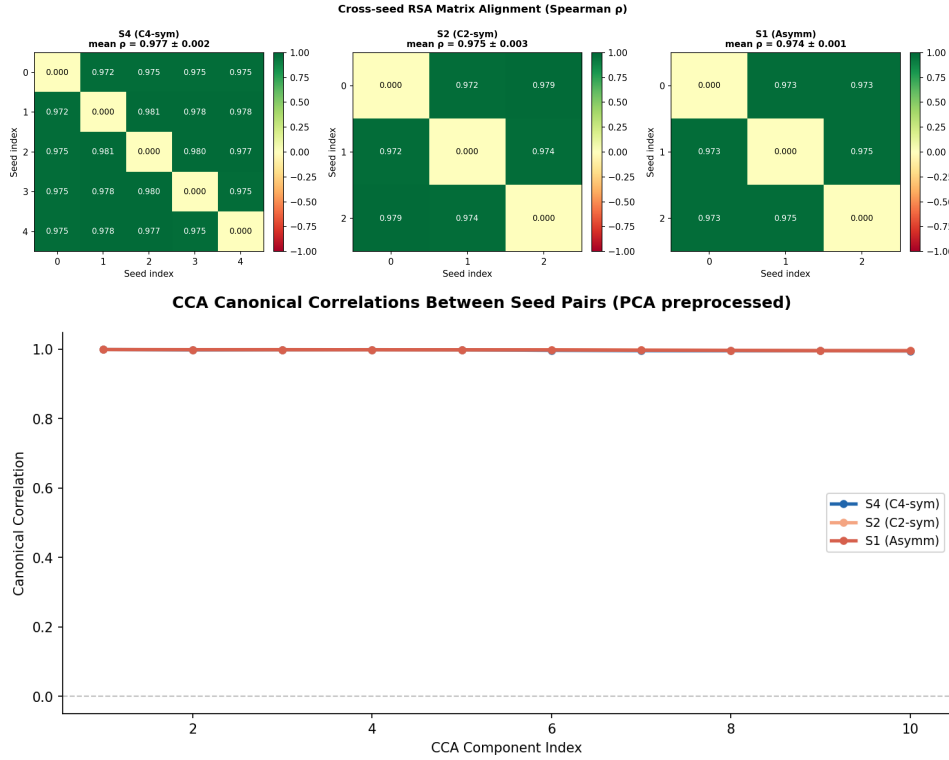


Figure 5: Cross-seed alignment analysis. Top: off-diagonal RSA values exceed 0.97 in all conditions. PAA gain does not improve upon identity alignment. Bottom: top canonical correlations remain near ceiling, indicating subspace alignment.

Even with C_4 -symmetric observations, the input stream itself is not symmetric: moving north from position (r, c) and from its 90° equivalent $(c, 17 - r)$ yields different head-direction-tagged transitions at different absolute orientations. These transitions remain globally consistent across any arena trajectory, so the network must track global heading precisely to minimize prediction loss. Global orientation therefore locks in deterministically at training initialization. This closes the first failure mode: global degeneracy is ruled out. These results are consistent with the Successor Representation interpretation [4]: representations are anchored by transition structure via HD-tagged state updates, which remain discriminable despite visual degeneracy. Whether this constitutes a direct proof of SR grounding (versus an alternative explanation) is left for future work. The next page shows that local precision costs cannot be prevented.

Unit-level aliasing scales monotonically with symmetry order

Head-direction input cannot prevent local aliasing costs. Unit-level rotational autocorrelation (RA) is S4 0.223 ± 0.003 , S2 0.116 ± 0.004 , S1 0.098 ± 0.002 —exactly matching symmetry group order. Shuffle z-scores are S4 15.07, S2 7.83, S1 6.48, all $p < 0.0001$; S4 versus S1 yields $p = 0.036$ and $r = -1.0$. Spatially tuned units (EVS > 0.93) comprise S4 1.8%, S2 3.7%, S1 4.7% of the population. Figure 6 shows the tuning distribution.

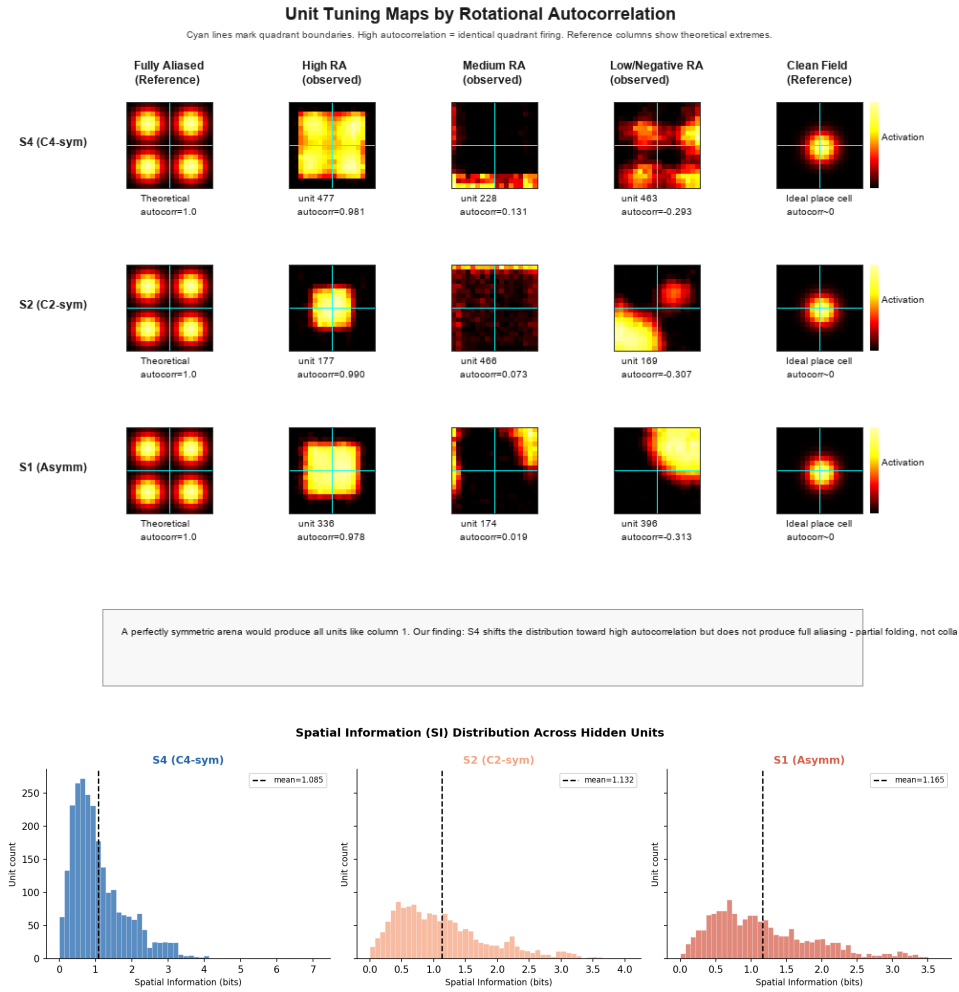


Figure 6: Unit-level tuning properties by symmetry. Top: rotational autocorrelation maps with quadrant boundaries marked. Red indicates high autocorrelation (cannot distinguish quadrants). Bottom: spatial information distributions with means indicated.

High-RA units in S4 produce four distinct peaks across quadrants—they encode within-quadrant position but cannot disambiguate which quadrant. Symmetric observations remove the prediction gradient that would force quadrant differentiation, so aliasing accumulates with symmetry order. Unit-level aliasing and population compression are separable.

Decoding degrades while population collapse stays partial

Population compression stays graded despite unit-level aliasing. Symmetry Collapse Index (SCI) is S4 0.889 ± 0.002 , S2 0.892 ± 0.000 , S1 0.984 ± 0.002 ; S4 versus S1 gives $p = 0.036$ and $r = -1.0$, but S4 versus S2 is not significant at $p = 0.786$. Linear decoding error worsens: S4 0.524 ± 0.014 , S2 0.475 ± 0.014 , S1 0.365 ± 0.012 ($p = 0.036$, $r = -1.0$). S4 compresses symmetry-related pairs by 11% relative to random pairs.

Figure ?? shows tuned cell fractions, SCI values, and decoding error. The dissociation between RA and SCI is striking: S4 and S2 show substantially different unit-level aliasing (RA: 0.223 versus 0.116) yet indistinguishable population compression ($p = 0.786$). The population code redistributes aliasing across units rather than concentrating it. The geometry of that compression changes under symmetry.

Distance geometry shifts toward path-dependence under symmetry

Symmetry transforms distance geometry. Distance-Topology Gap (DTG) is $S4 -0.052 \pm 0.003$, $S2 -0.011 \pm 0.003$, $S1 -0.020 \pm 0.004$ ($p = 0.036$, $r = 1.0$). All gaps are negative, indicating city-block dominance, but the gap widens in S4. S2 DTG of -0.011 ± 0.003 is unexpectedly similar to S1 (difference not significant, $p = 0.197$).

Figure 7 shows how distance-topology gap diverges during training.

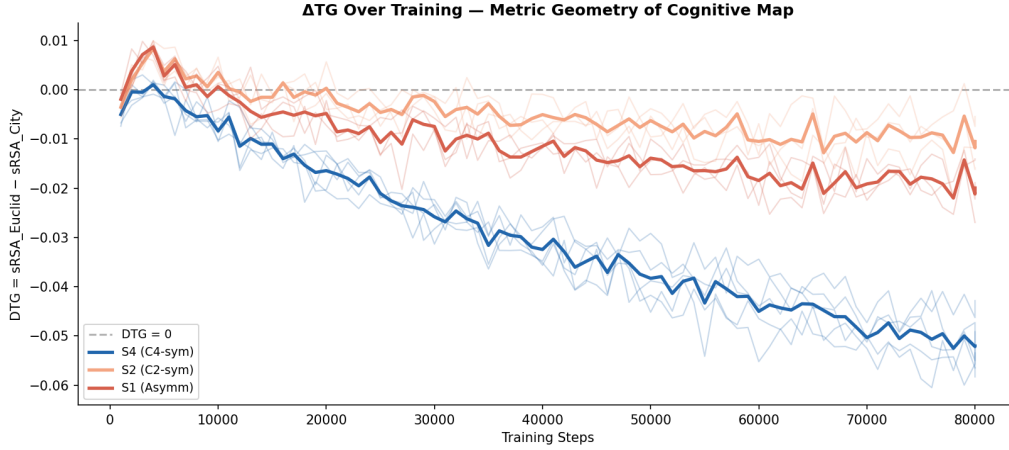


Figure 7: Distance-Topology Gap (Δ_{TG}) over training. Negative values indicate city-block structure dominance. Gap widens (becomes more negative) in S4.

Under C_4 symmetry, inter-quadrant Euclidean structure collapses because identical observations push those positions toward similar hidden states; path length within quadrants dominates the learned distance geometry. The 18×18 MiniGrid imposes a city-block navigation metric by construction—agents move only along grid lines—so this result requires replication in a continuous environment (e.g., RatInABox) before representational geometry can be separated from environmental structure.

C2 symmetry is encoded as group structure in the population code

The population code encodes group structure, not all symmetries uniformly. C2 Contrast is $S2 +0.045 \pm 0.003$, $S4 -0.130 \pm 0.002$, $S1 -0.070 \pm 0.004$ ($p = 0.036$ S4 vs S2, $p = 0.100$ S2 vs S1). In S2, 180° -related positions are more similar to each other than 90° -related—the code groups by C_2 subgroup. In S4, all quadrants are equally confusable, so inter-quadrant pairs compress identically and subgroup structure vanishes.

The network learned this symmetry group structure from prediction error alone, without explicit group objectives. Observation equivalence structure is directly reflected in representational similarity structure—a result linking algebra to neural coding. Results pages 6 through 10 establish that symmetry costs are local, graded, measurable, and architecturally constrained.

Manifold geometry stays high-dimensional and folds along symmetry axes

The network maintains high dimensionality despite symmetry. TwoNN intrinsic dimensionality is S4 5.95 ± 0.14 , S2 5.78 ± 0.23 , S1 5.66 ± 0.26 .

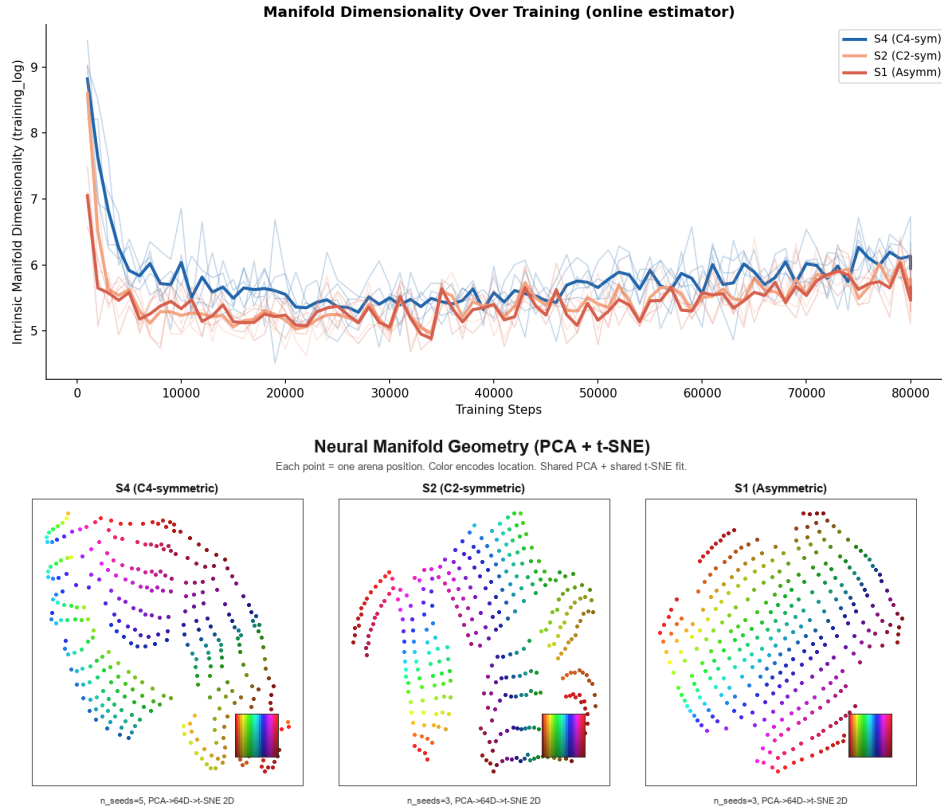


Figure 8: Manifold geometry by symmetry condition. Top: online intrinsic dimensionality estimates during training converge to 5.7–6.0. Bottom: t-SNE visualizations (PCA to 64D first). S1 shows continuous gradient; S4 shows fragmented multi-arm structure; S2 shows symmetric two-lobe structure.

The manifold fragments locally while preserving global structure—partial folding. The network cannot fold symmetric positions onto a single point without destroying within-quadrant structure for multi-step prediction, so it maintains high dimensionality while tolerating inter-quadrant ambiguity. Across all results: global degeneracy is ruled out, local aliasing scales with symmetry, population compression stays graded, distance geometry shifts toward path-dependence, the population code encodes group structure, and the manifold stays high-dimensional but fragments. Symmetry costs are real but bounded and compensable.

Predictions the model generates for hippocampal biology

The model predicts that multi-field place cell fraction should scale with symmetry group order. This follows directly from the RA result: existing hippocampal recording datasets from symmetric arenas can be stratified by symmetry order and the fraction of multi-field units measured. The RA result predicts that multi-field fraction scales as $|C_4| > |C_2| > |C_1|$, matching the symmetry groups: RA = 0.223, 0.116, 0.098 respectively. This prediction follows from the computational finding that high-RA units respond identically at rotationally equivalent positions and cannot disambiguate arena quadrant—precisely the dual-frequency structure of multi-field place cells recorded in symmetric arenas [3, 1].

The model generates a specific prediction about head-direction lesion effects. Lesioning anterior thalamic nuclei (which provide HD input to hippocampus) should raise PAA gain above 0.05 in S4 arenas while RA remains at 0.223 or higher. This dissociation is the key prediction: global orientation becomes ambiguous between seeds (PAA breaks) but local aliasing persists (RA unchanged), because HD input stabilises global phase through transition statistics without affecting local observation equivalence. This dissociation is unique to the partial folding account.

The DTG result generates a testable prediction about population vector geometry. In symmetric arenas, population vector distances should correlate more strongly with path-length than Euclidean distance, with the correlation bias largest in C_4 arenas and intermediate in C_2 arenas. The DTG result quantifies this: city-block bias is -0.052 ± 0.003 in S4 versus -0.020 ± 0.004 in S1. This prediction is directly testable in existing hippocampal recording data.

Limitations and replication requirements

Sample sizes are at the statistical minimum. The sample sizes are $S4\ n = 5$, $S2\ n = 3$, $S1\ n = 3$, giving minimum uncorrected p -values of 0.036 and minimum Bonferroni-corrected p -values of 0.107 (correction factor = 3 comparisons). All main comparisons sit at theoretical minimum uncorrected p and maximum effect size ($r = -1.0$), indicating the effects are as large as statistically observable at these sample sizes. Replication requires $n \geq 8$ seeds per condition to achieve corrected significance ($p < 0.05$ after Bonferroni correction for 3 metrics).

The SCI and RA dissociation is unexplained. $S4$ and $S2$ show substantially different unit-level aliasing (RA: 0.223 ± 0.003 versus 0.116 ± 0.004) yet statistically indistinguishable population compression (SCI: $p = 0.786$). This dissociation suggests the population code redistributes aliasing across units rather than concentrating it through compensatory coding strategies. The mechanistic cause remains unidentified and requires analysis of learned weight matrices and representational geometry.

Distance geometry results confound representation with environment. The 18×18 MiniGrid imposes a city-block navigation metric by construction—agents must move along grid lines. The DTG result showing larger negative gaps under symmetry may partially reflect this environmental structure. Replication in a continuous environment such as RatInABox is required before the representational geometry interpretation is secure.

Spatially tuned unit fraction is lower than typical recording values. The fraction of spatially tuned units at $EVS > 0.93$ is $S4\ 1.8\%$, $S2\ 3.7\%$, $S1\ 4.7\%$, substantially lower than values typically reported in hippocampal recordings at more lenient thresholds ($EVS > 0.5$). The more conservative threshold reflects position-averaged evaluation which removes within-position variability. The ordering $S4 < S2 < S1$ is robust to threshold choice and the main conclusions rest on population-level metrics (RA, SCI, DTG, C2 Contrast) unaffected by this discrepancy.

Conclusion

Symmetric landmarks do not destroy predictive maps—they impose a specific cost structure. Head-direction input prevents global orientational degeneracy through oriented transition statistics that anchor the global phase of the attractor, but landmark symmetry degrades local precision monotonically with symmetry group order. This dissociation emerges mechanistically: the transition function encodes directional information unambiguously even under symmetric observations, so the global frame stays locked in, while symmetric observations remove the prediction gradient that would push units to differentiate rotationally equivalent positions locally. The result is partial folding: stable global orientation paired with graded local compression.

PAA detects global degeneracy, RA detects unit-level aliasing, and C2 Contrast detects group structure encoding. These three metrics do what sRSA cannot: separate identifiability from entanglement from population organization. sRSA remains high while local precision collapses under symmetry because it is invariant to permutations. This framework applies to any predictive system facing observation aliasing—not just spatial maps, but developmental learning and abstract state spaces where observations are non-invertible.

The dissociation between global stability and local precision is not a failure of predictive learning—it is a mechanistically interpretable consequence of the transition structure available to the learner. HD-anchored transitions stabilise global phase, but they cannot resolve local ambiguity created by observation aliasing. Future work should vary the oriented context signal directly: remove HD input entirely, degrade its signal-to-noise ratio, or replace it with a learned orientation signal. These manipulations will test how much of the global stability result depends on the specific form of the HD input versus the general principle—and the three metrics introduced here are ready to measure the answer in any future extension of the Levenstein framework, including continuous environments, developmental training regimes, and non-spatial state spaces.

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